MILITARY SPECIFICATION

POWER, DISTRIBUTION PANEL, ALTERNATING CURRENT, F-4 AIRCRAFT, SPECIFICATION FOR

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 This specification covers the requiemendts for an alternating current power distribution panel for installation In military aircraft aircraft. The panel shall be for use In 30KVA 400 cps, ll5/200 volts, 4 wire. grounded neutral. we reconnected, a-c electric systems in accordance with MS90302. This panel is designated by Military part Number M81305-1.

2. APPLICABLE DOCUMENTS

2.1 Issues The following documents, of the issues effect on the date of the invitations for bids, are cited by this specification for certain requirements. Whenever my requirement of any document cited by this specification conflicts with any requirement explicitly stated in this specification, the re-quirement explicitly stated In this specification shall prevail.

SPECIFICATIONS

Military

MIL-M-14	Molding Plastics and Molded Plastic Parts, Thermosetting
MIL-P-997	Plastic-Material, Laminated, Thermosetting, Electrical-Insulating: Sheets, Glass Cloth, Silicone Resin
MIL-C-5015	Connectors, Electrical, “AN” Type
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-R-6106	Relays, Electric, Aerospace, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series; General Specification for
MIL-T-8654	Transformer, Current, Aircraft
MIL-P-15037	Plastic sheet, laminated, Thermosetting Glass Cloth, Melamine-Resin
MIL-P-17555	Electronic and Electrical Equipment and Associate Repair Parts, Preparation for Delivery

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MIL-G-21480	Generator System, Single Generator, Constant Fequency Alternating Current Aircraft, Class C, General Specification for
MIL-W-22759	Wire, Electric, Fluorocabon Insulated, Copper
MIL-C-45662	Calibration System Requirements
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-104	Limits for Electrical Insulation Color
MIL-STD-129	Marking for Shipment and Storoge
MIL-STD-130	identification Marking of U.S. Military Property
MIL-STD-143	Specification and Standards, Order of Precedence for the Selection of
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-456	Electronic Parts, Date and Source Coding for

DRAWINGS

Military

MS90302	Generator Systems, Alternating Current, Brushless, 30KVA

2.2 Other Publications - The followlng document forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for request for proposal shall apply:

WR-62 Naval Weapons reqauirements, specifications and standards; Use of.

3. REQUIREMENTS

3.1 The distribution panel furnished under this specification shall be a product which has been tested, and has passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Dimensions - The dimensions of each panel conform to Figure 1.

3.3 Weight - The weight of each panel shall not exceed 38 pounds.

3.4 Materials - Materials shall conform to applicable specifications and shall be as specified herein. Materials which are not covered buy applicable specifications or which are not specifically described herein shall be of the best quality, of the lightest possible weight, and suitable for the purpose intended and shall be selected in accordance with Navel Weapons Requirement WR-62.

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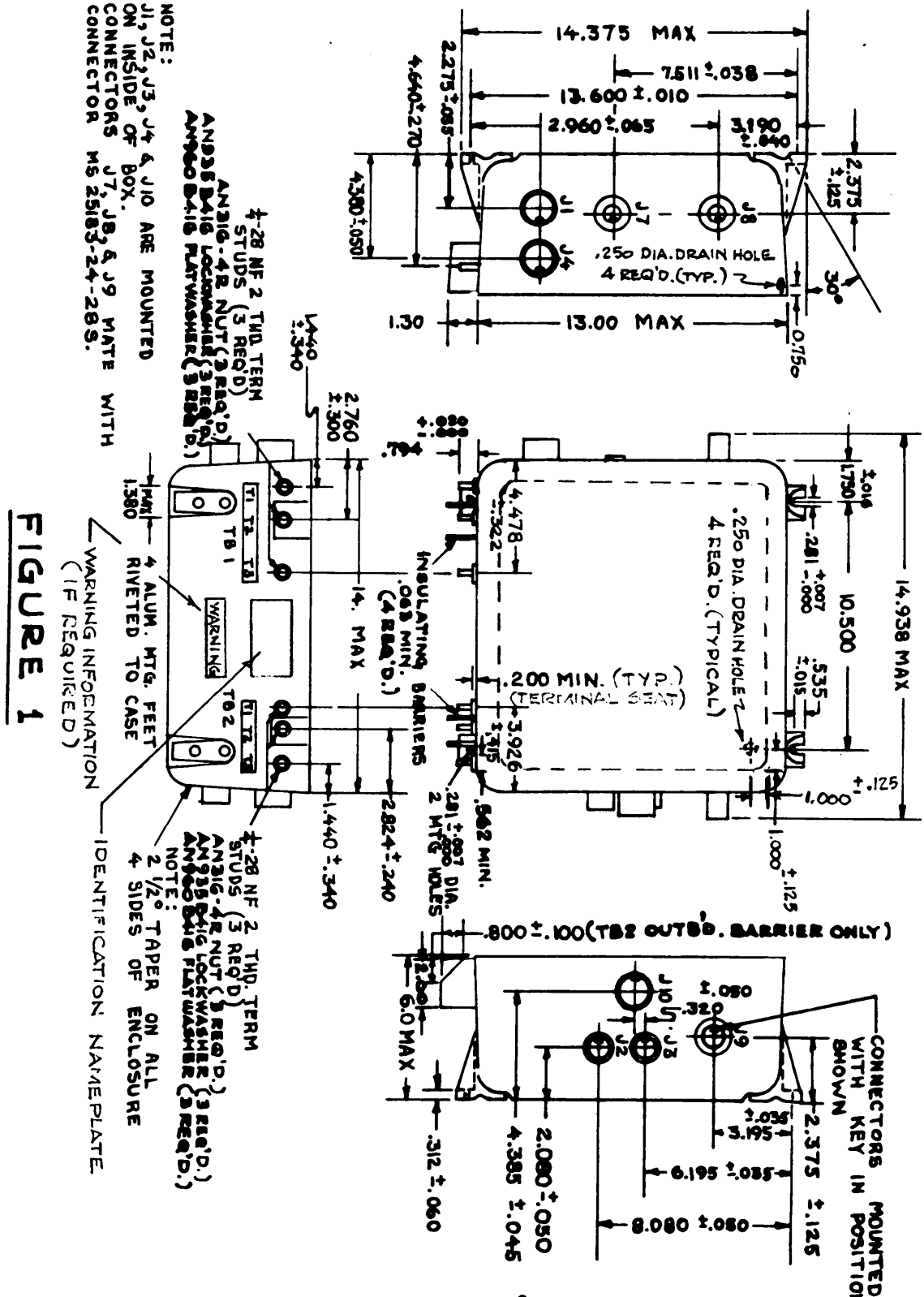


FIGURE 1

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3.4.1 Metals - Metals shall be of a corrosion-resistant type or shall be suitably processed to resist corrosion. The use of dissimilar materials shall comply with MIL-STD-454 Requirement 16. This paragraph does not prohibit the use of dissimilar metals within hermetically sealed enclosures.

3.4.2 Nonmetals - For external parts of sealed components and all parts of unsealed components, nonmetals, including plastics, fabrics and protective finishes shall be moisture flame resistant, shall not support fungus growth, and shall not be adversely affected by weathering and aircraft fluids, at the temperture specified.

3.4.2.1 Plastic Material - Except as specified herein, plastic material shall conform to Specification MIL-M-14 for molded material and Specification MIL-P-997 or Mil-P-15037 for laminated material. Other types of plastic materials possessing superior charteristics may be used, provided the manufacturer submits acceptable evidence to the activity responsible for qualification that such material meets as applicable the performance requirements of Specification MIL-M-14, 4XL-P-997, or MIL-P-15037. In addition to these specifications and the requirements of 3.4.2, the plastic materials shall not support combustion, give off noxious gases in harmful quantities, give off any gases in quantities sufficient to cause tracking between current carrying parts or cause contamination of contacts.

3.4.3 Selection of Materials - Specifications and standards for all materials, parts, and equipment not specifically designated herein, which are necessary for the execution of this specification, shall be selected in accordance with Naval Weapons Requirement WR-62 except as provided in the following paragraph:

3.4.3.1 Standard Parts - Standard parts (MS or AN) shall be used wherever practicable, and shall be identified on the contractor's drawing by their military part numbers. Commercial utility parts such as screws, bolts, nuts, and cotter pins may be used, provided they possess suitable properties and are replaceable by the standard parts (MS or AN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial may be used provided they conform to all requirements of this specification.

3.5 Design and Construction - The panel design shall not exceed the envelope dimensions shown on Figure 1 and shall meet the mechanical shocks the stresses specified herein. The panel shall be designed to operate in aircraft electrical systems conforming to the requirements of MS90302.

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3.5.1 Housing - The panel housing shall be constructed of fiber glass or equivalent nonmetallic material in accordance with Figure 1.

3.5.1.1 Interior Preparation - Before installation of components the interior of the panel housing shall be white in accordance with MIL-STD-l04, class 1.

3.5.1.2 Attitude - The panel shall be constructed to Insure proper operation when mounted in any position.

3.5.2 Terminal studs - The panel shall have electric terminal studs as shown in Figure 1. The terminal studs shall be designed to withstand for 1 minute, without damage, the static values of pull and torque specified in MIL-R-6106.

3.5.2.1 Design - Terminal studs shall be of threaded design to allow the securing of 3 MS25036-20 terminal. lugs with a minium of one complete thread exposed above the applicable nut adequately torqued on the 3 lugs.

3.5.2.3 Finish - Finishes applied to the surfaces of the terminal stud shall. meet all perfomance requirements specified herein.

3.5.2.4 Terminal Stud Barriers - The panel shall be equipped with terminal stud barriers in accordance with Figure 1 to separate the terminal studs and lugs and protect against inadvertent shorting, grounding or contact by personnel. The barriers shall meet all the performance requirements of this specification.

3.5.3 Electric Receptacles - Electric receptacles shall comply with the requirements of MIL-C-5015 and the tests of Section 4 of this specification when mated with the corresponding plugs.

3.5.4 Marking - Electric connectors and terminals shall be durably and legibly identified as shown on Figure 1.

3.5.5 Wire Marking - All electric hook-up wire shall be marked at intervals not greater than 6 inches and adjacent to the terminations so that they may be readily identified without disturbing wire bundles or disconnecting wires from any termination. The markings shall agree with the terminal marking data shown on the manufacturer's wiring diagram.

3.5.6 Screw Threads - Screw threads shall be in accordance with Specification MIL-S-7742.

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3.5.7 Mounting - Mounting shall be as shown In Figure 1.

3.5.8 Clearance - Adequate clearance shall be provided for suitable terminations and mounting hardware. No special installation tools shall be necessary for service or maintenance of the panel.

3.6 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to Installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-70327.

3.7 Performance - The panel shall meet the Quallty Assurance Provisions of Section 4 specified herein.

3.8 Relays - Unless otherwise specified herein all relays and contractors used in he panel shall be hermetically sealed and shall meet the requirements of MIL-R-6106. They shall be tested and meet the requirements of sampling Plan A and B of MIL-R-6106 and meet the rating and operating characteristics specified in paragraph 3.8.1 and 3.8.2 of this specification. The Left Line Contactor (LLC), Right Une Contractor (RIC) and Tie Contactor (TC) shall consist of a three-pole, single-throw, normally open set of main contacts, and six normally open and six normally closed single-throw auxiliary contacts. The External Power Contactor (EPC) shall consist of a three-pole, single-throw, normally open set of main contacts, and one normally open and one normally closed single-throw auxilary contact. The Loop Shorting Relay (LSR) sha11 consist of at least three normally closed single-throw contacts. The auxillary contacts shall be completely isolated from the main contacts so that there shall be no contamination of the auxillary contacts from arcing of the main contacts. The main and auxiliary contacts shall be operated by a common actuator. The minimum current requirements of MIL-R-6106 shall be applicable to the LSR and all auxillary contacts.

3.8.1 Contact Rating - The rating of the main (power) contacts shall be at least 120 amperes resistive and 120 amperes inductive at 115/200 volts 400 cps. The rating of the auxiliary contacts shall be at least 5 amperes resistive and 3 amperes inductive at 28 volts d-c. The LSR contacts shall be rated at least 5 amperes resistive and 3 amperes inductive at 10 volts 400 cps.

3.8.2 Operating Characteristics -

3.8.2.1 Operating Time - The operating or pickup time of the main contacts of LLC, RLC, TC, and EPC shall be 70 milliseconds maximum at maximum pickup voltage at 25°C ±50.

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3.8.2.2 Release Time - The release or drop-out time of the main LLC, RLC, TC, and EPC contacts shall be 20 milliseconds maximum.

3.8.2.3 Maximum Coil Operating Voltage - . 30 volts dc, all relays contractors.

3.8.2.4 Maximum Pickup Voltage - 18 volts dc, max., all relays and contractors.

3.8.2.5 Maximum Drop-Out Voltage - 7 volts dc, max., all relays and contractors.

3.8.3 Paralleling - During generator paralleling the 3 current transformer loops shall remain shorted 5 to 20 milliseconds after the TC has closed.

3.9 Current Transformers - All current transformers (CT) shall meet the requirements of MIL-T-8654, the performance requirements of MS90302 and the following:

(A) Type	Inserted Primary
(B) Rating	250 Amperes
(C) Design Ratio (with primary cable passing through the aperature once)	125:1
(D) Frequency Range	350 to 450 CPs
(E) Burden Impedance (Rtjx) @400 CPS	13.3 +j7.4 ± 10 ohms
(F) Error (between 25 and 111 Amps.)	
Phase Angle	less than 2°
Ratio	less than 5%

3.10 Electrical Protection - The panel shall have an insulated enclosure provided with drain holes in accordance with Figure 1. All components in the panel shall. be isolated from each other and from the panel mounting. All externally exposed screws, screw heads, etc., securing internal components to the panel, shall be coated with a suitable insulating compound.

3.11 Wiring - Internal wiring of the a-c power distribution panel shall conform to the requirements *of* Figure 2.

3.11.1 Wiring Diagram - A diagam showing the wiring connections shall be located on the interior of each panel.

3.12 Identification of Product - The panel shall be marked In accordance with MIL-STD-130 and shall include the following information:

Name of Unit	30 KVA, 400 cps 115/200 volts
Federal Stock Number	Date and Source Code
Manufacturer's Part No.	(in accordance with MIL-STD-456)
Manufacturer's Serial Mo.	Manufacturer's Name
	Military Part Number M81305-1

3.12.1 MIL Designations - MXL designation shall not be applied to a product except for qualification test samples until notification has been received that the product has been approved by the Qualifying Activity.

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NOTES: ALL CONTROL WIRES SHALL BE #22 AWG UNL.
ALL POWER WIRES SHALL BE #6 A
BUS-BAR UNLESS OTHERWISE NOT
① TO LEFT GENERATOR WARNING LIGH
② TO TIE CONTACTOR WARNING LIG
③ TO RIGHT GENERATOR WARNING LI
④ TO FREQUENCY AND LOAD CONTROL
⑤ CONTROLS TC WHEN OPERATING ON
⑥ TO EXTERNAL POWER RECEPTACLE TH
⑦ TO EXTERNAL POWER SWITCH AND EMERG
⑧ CONTROLS EMERGENCY GEN
⑨ TO EMERGENCY GENERATOR R

TB1 & TB2 TERMINAL BOARDS (REF. FIG. 1)

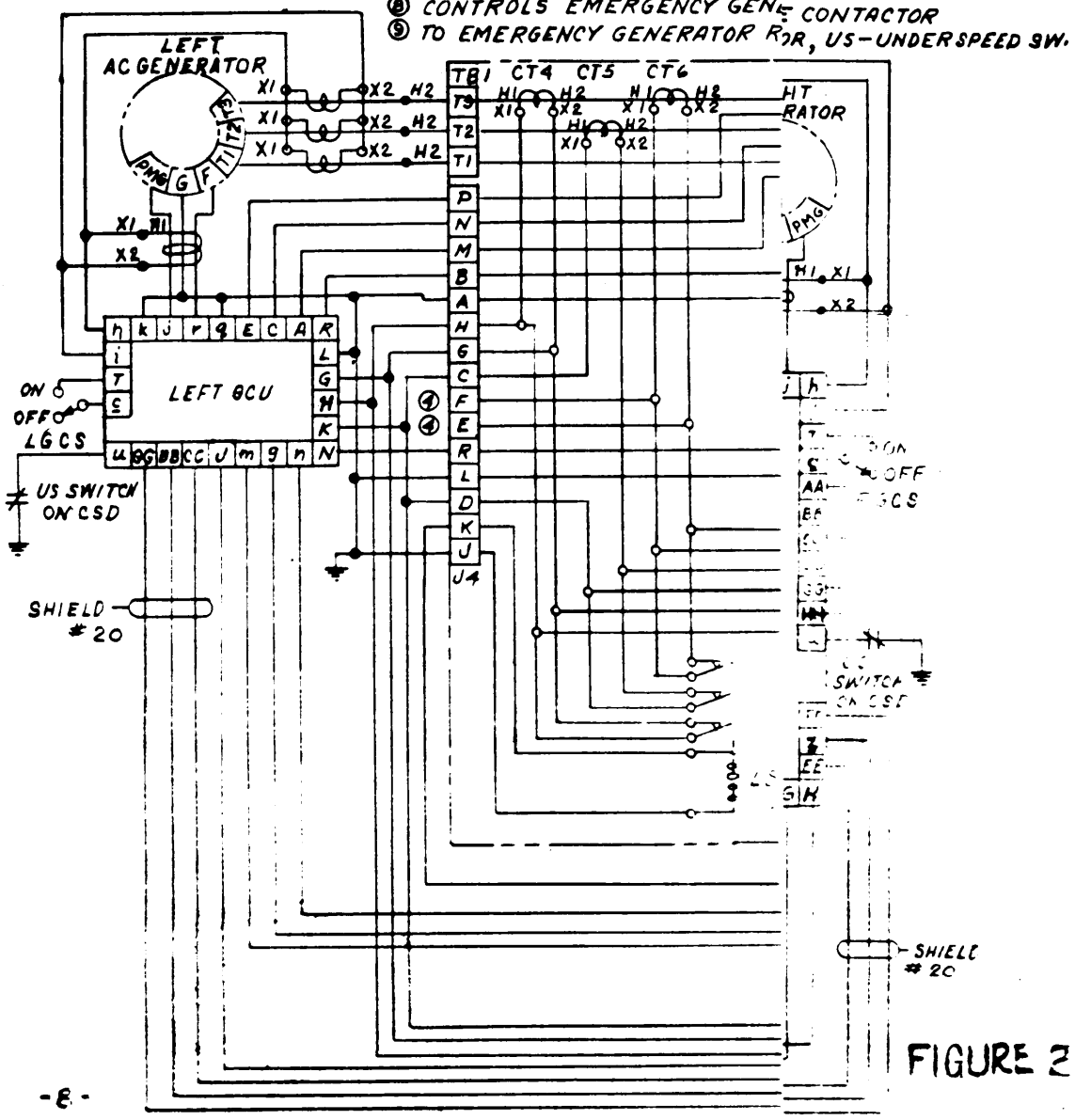


FIGURE 2

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3.13 Workmanship- Workmanship shall be in accordance with the best commercial manufacturing practice covering this class of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Quality Assurance - Unless otherwise specified in the contract or purchase order, the manufacturer is responsible for the performance of all quality assurance tests specified herein. Upon notification of qualification approval no changes to the units being produced, affecting the performance or interchangeability of the equipment, shall be permissible without prior approval by the government activity having responsibility for qualification (6.3). Except as otherwise specified the mufacturer may use his own facilities or any commercial laboratory acceptable to the Government. The government reserves the right to perform any of the tests set forth in this specification where such tests are deemed necessary to assure that supplies and services conform to prescribed requirements. Records of the examinations and tests shall be kept complete and available to the Government.

4.2 Classification - The inspection and testing of the panels shall. be classified as follows:

(a) Qualification Tests: (4.3)
(b) Quality conformance tests: (4.4)

4.3 Qualification Tests

4.3.1 Sampling Instructions - Qualification test samples shall consist of four panels upon which qualification is desired. These panels shall be accompanied by one reproducible and two copies of schemstic and wiring diagrams, outllne and detail assembly drawings, and two copies of instructions with illustrations and diagrams, covering installation of the panel. Samples shall be forwarded to the Qualitying Activity (6.3) plainly identified by securely attached durable tags marked with the following information (as applicable):

Sample for Qualification Test
Panel - A-C Power Distribution; Aircraft
Name of Manufacturer
Manufacturer's Part Number (P/N) and Serial Number (S/N)
Military Part Number M81305-1
Submitted by (name) (date) for qualification test in accordance with the requirements of Specification MIL-P-81305(WP) under authorization (reference authorizing letter).

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4.3.2 Tests - The qualification tests shall consist of all the tests of this specification. These tests shall be performed on the three panels in accordance with the schedule and order shown in Table I.

TABLE I

TESTING SCHEDULE

Test	Test Paragraph	Sample Number 1 2 3 4
Examination of product	4.6.1	X X X
Strength of terminals	4.6.2	X X X
Dielectric strength	4.6.3	X X X
Temperature-altitude dielectric	4.6.4	X X X
Circuit continuity	4.6.5	X X X
Voltage drop	4.6.6	X X X
High Current (Power) Circuits	4.6.6.1	X X X
Low Current (Control) Circuits	4.6.6.2	X X X
Current transformer accuracy	4.6.7.1	X
Current transformer tracking	4.6.7.2	X
Simultaneous operation on pickup	4.6.8	X
Simultaneous operation on cropout	4.6.9	X
Functional test	4.6.10	X X X
Thermal shock	4.6.11	X
Acceleration	4.6.12	X
Vibration	4.6.13	X
Shock	4.6.14	X
Salt Spray	4.6.15	X
Humidity	4.6.16	X
Vibration scan	4.6.17	X
Fungus	4.6.18	X
Inspection	4.3.2.1	X X X
Flight Test (100 hours)	4.6.19	X

4.3.2.1 Inspection - At the conclusion of the tests, the panels shall be inspected for excessive wear, defects, and damage of any type.

4.4 Quality Conformance Tests - Quality conformance tests shall consist of sampling plans A, B, and C tests.

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4.4.1 Sampling Plan A Tests - Each panel shall be subjected to the sampling plan A tests which consist of the tests specified in Table II.

TABLE II

SAMPLINGP PLAN A TESTS

Test	Paragraph
Examination of product	4.6.1
Dielectric strength (at sea level)	4.6.3
Circuit continuity	4.6.5
Functional test	4.6.10

4.4.1.2 Rejection and Retest - Units that have been rejected may be reworked or have parts replaced to correct the defects and then be resubmitted for acceptance. Units rejected after retest shall not be resubmitted for testing without the specific approval of the Government Inspector.

4.4.2 Sampling Plan B Tests

4.4.2.1 Sampling Procedure - Under sampling plan B tests, 2 panels shall be selected at random from every 100 units of each manufacturer's part number produced. Sample units that have been subjected to and have passed samplng plan A tests shall be used, unless the Government considers it more practical to select separate samples from the lot for sampling plan B tests.

4.4.2.2 Tests - Sampling plan B tests shall consist of the test specified In Table III and performed in the order shown. Sampling Plan B panels may be shipped upon satisfactory completion of these tests.

TABLE III

SAMPLING PIAN B TESTS

Test	Paragraph
Temperature-altitude dielectric	4.6.4
Circuit voltage drop (DC only)	4.6.6
Simultaneous operation on pickup	4.6.8
Simultaneous operation on dropout	4.6.9
Vibration scan	4.6.17

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4.4.2.3 Rejection and Retest. - When one or more items fai1 to meet the specification, further acceptance will be withheld until the failures have been corrected. After corrections have been made, all necessary tests shall be repeated.

4.4.2.4 Sampling Plan A Tests Continuance. - For production purposes, sampling plan A tests may be continued pending the investigation of a sampling plan B test failure. Final acceptance of the entire lot, however, shall not be made until it is determined that the lot meets all the requirements of the specification.

4.4.3 Sampling Plan C Tests

4.4.3.1 Sampling Procedures. - Unless otherwise specified by the procuring activity , 2 panels shall be selected at random from every 500 units of each manufacturer's part number produced.

4.4.3.2 Tests. - Sampling plan C tests shall consist of all tests specified in Table I, with the following exceptions:

Acceleration	(4.6.12)
Vibration	(4.6.13)
Salt spray	(4.6.15)
Humidity	(4.6.16)
Fungus	(4.6.18)

4.4.3.3 Rejection. - When one or more items selected from a production run fail to meet the specification, no items still on hand or later produced shall be accepted until the failures-have been corrected.

4.4.3.4 Sampling Plan A Tests Continuance - For operational purposes, sampling plan A tests may be continued pending the investigation of a sampling plan C test failure. Final acceptance of items on hand or produced later shall not be made, however, until it is determined that items meet all the requirements of the specification.

4.4.4 Defects in Items Already Accepted - The investigation of a test failure could indicate that defects may exist In items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.5 Test conditions - Unless otherwise specified, the following test conditions shall be used.

4.5.1 Temperature - The ambient temperature shall be 25° ± 10°C.

4.5.2 Pressure. - The ambient pressure shall be that pressure prevailing at the the and place of the test (normally approximately sea-level altitude).

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4.5.3 Humidity - The relative humidity shall not be greater than 80 per cent.

4.5.4 Power Supply - The test power supply shall be capable of simulating the MIL-G-21480 power conditions in such a way that a-c voltages shall be within the steady-state and transient voltage limits specified in MS90302.

4.6 Test Methods

4.6.1 Examination of Product - Each panel shall be inspected to determine conformance with this specification and all applicable drawings with respect to material, standard parts, simplification, electrical connections, threaded parts, finish, interchangeability, instructions, marking, dimensions, weight, and overall workmanship.

4.6.2 Strength of Terminals - The specified pull force shall be applied both coaxially with the threaded terminal in a direction away from the main body of the a-c power distribution panel, and again normal to the threaded axis of the terminal in approximately the same plane as the seat for the terminal lug. The specified torque shall be applied to the terminal mounting nut, or screw, with all terminal mounting hardware, including one terminal lug of proper size assembled in proper order.

4.6.3 Dielectric Strength - The panel shall withstand the dielectric test voltages (rms) at 60 cycles per second, between mutually insulated parts, and also between current-carrying parts and all noncommon exposed and grounded metal parts, specified in Table II under paragraph 3.5.2 of MIL-R-6106E. These tests shall be made with the relays in both the energized and deenergized positions. Any arc over breakdown or current flow which constitutes a transient or steady increase in leakage current at the specified test voltage shall constitute failure.

4.6.4 Altitude-Temperature - The panel shall be brought to a stabilized condition at each of the altitude temperature conditions specified in Table IV. The tests of 4.6.3 shall then be performed. The altitude test potential of MIL-R-6106 shall be used for the dielectric strength test.

TABLE IV

Pressure Altitude Thousands of Feet	Continuous °C	30 Minutes °C	4 Minutes °C
0	72	95	102
30	55	80	140
70	--	--	70

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Test Procedure: (All temperatures listed are chamber air temperatures external to the equipment).

(1) Maintain the continuous operating temperature until the equipment has reached a stable temperature.

(2) Increase the temperature to the values specified under the 30 minutes column and hold for a period of 30 minutes.

(3) Reduce the temperature to the continuous value and maintain this value until the equipment has reached a stable temperature.

(4) Increase the temperature to the values specified under the 4 minutes column and hold for a period of four minutes.

4.6.5 Circuit Continuity - Each individual circuit of the panel shall be checked by making a point-to-point test with an ohmmeter or other suitable method acceptable to-the procuring agency to determine conformance to the circuit diagram. This test shall be made with the relays first de-energized and then energized to insure that both normally closed (NC) and normally open (NO) contacts are checked. Each circuit checked sha1l Include all relay contacts and connector pins pertinent to that circuit. All NC contacts shall remain closed in the deenergized position and all NO contacts shall be closed in the energized position. No NC and NO contacts shall be made at the same time. The check list shall be kept as a permanent record available to the Government inspector or qualifyng activity.

4.6.6 Voltage Drop - With the circuit carrying rated current, at the specified voltage, the circuit shall be caused to break and then make the current before each of three consecutive measurements made within 10 seconds after the contacts close. Coil voltage shall be as specified in paragraph 3.8.2.3.

4.6.6.1 High Current (Power) Circuits - The maximum average voltage drop across each pair of designated terminals shall not exceed the following:

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CIRCUIT	SOURCE VOLTAGE 28 volts DC	115 Volts AC 400 CPS
Ran the left hand generator input terminals to the left hand bus terminals.	175 MV	500 MV
From the right hand generator input terminals to the right hand bus terminals.	175 MV	500 MV
From the left hand generator input terminals to the right hand bus terminals with the TC energized.	325 MV	500 MV
From the right hand generator input terminals to the left hand bus terminals with the TC energized.	325 MV	500 MV
From the external power receptacle (J3) to the left hand bus terminals with the TC energized.	325 MV	500 MV
From the external power receptacle (J3) to the right hand bus terminals with the TC energized.	325 MV	500 MV

4.6.6.2 Low Current (Control) Circuits - The d-c resisance of control circuits shall not be greater than 0.2 ohm.

4.6.7 Current Transformer -

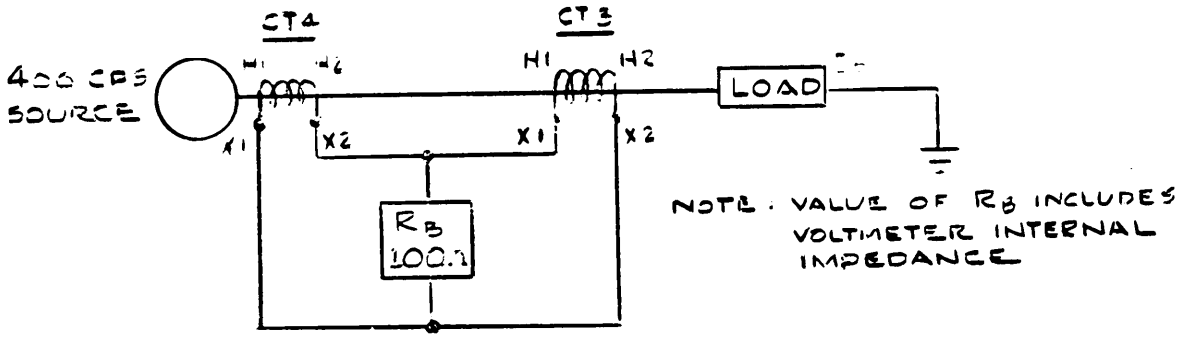
4.6.7.1 Accuracy - Accuracy of the current transformer (CT) shall be measured at 350, 400 and 450 cps with a burden as specified in 3.9 and a line power factor of 0.75 with a sine wave current of 25 to 111 amperes flowing through the power line. The ratio and phase angle errors shall not be greater than the values specified in 3.9. When calculating the phase angle errors, the error introduced in the measuring instruments, due to the ratio error of the CT under test, shall be factored out. The ratio error shall be calculated by the following formula and shall be positive in algebraic sign:

$$\text{Percent ratio error} = \frac{(\text{measured current ratio}) - (\text{design ratio})}{(\text{design ratio})} \times 100$$

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4.6.7.2 Tracking - Transformers 3 and 4 shall be connected series aiding as shown in Figure 3. The load wire shall be fastened so as to be in the same re1ative position in both transformers. The primary load current at 400 cps shall be varied from no load to fill load in convenient steps. The voltage drop across R_B shall not be greater than .200 with load currents (I_P) from 0 to 125 amperes. This test shall be repeated for transformer pairs 2 and 5 and for 1 and 6.

FIGURE 3



CURRENT TRANSFORMER TEST

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4.6.8 Simultaneous Operation on Pickup - Nominal operation voltage shall be applied to each relay and contactor coil in turn. All NO contacts on any one relay or contactor shall operate simultaneousIy within 5 milliseconds.

4.6.9 Simultaneous Operation on Dropout - Nominal operating voltage shall be removed from each relay and contactor coil in turn. All NC contacts on a given relay or contactor shall operate simultaneously (6.4.11) within 10 milliseconds.

4.6.10 Functional Tests - The panel shall be connected to an electric circuit which simulates the actual aircraft electric operation. The circuit shall monitor for correct circuitry and phase relationship. Bus load shal1 be 15 ±5 Amps. at 115/200 volts, 3 phases a-c. At the manufacturer's option, this test may be run with indicating lamps at a maximum of 115 volts a-c. The panel on test shall be operated to simulate the following functions:

A. All relays and contractors shall be checked for pickup and dropout voltage at maximum temperature or a calculated equivalent pickup and dropout voltage lt room temperature.
B. Energize the tie contactor (TC) and energize the external power contactor. Check for power on the left-hand AC bus and right-hand AC bus. De-energize tie contact or and external power contactor Power shall drop to zero on both buses.
C. Energize left line contactor (LLC). Power shall be on left hand AC bus. De-energize (LLC). Power shall drop to zero.
D. Energize right line contactor (RLC). Power shall be on right hand AC bus. De-energize (RLC). Power shall drop to zero.
E. Cycle all relay and contactor coils without electrical load 25 times, and check circuit continuity on all contacts. Rerun “B", “C”, and “D”.
F. Each pair of current transformers shall meet the requirements of par. 4.6.7.

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4.6.11 Thermal Shock - The panel shall be subjected to thermal shock test of MIL-STD-202, test condition B. Upon completion of this test the panel shall successfully complete the functional test, 4.6.10.

4.6.12 Acceleration - The panel shall be subjected to the acceleration test Procedure III of MIL-E-5272 except that the acceleration force shall be 15 g in each axis in both directions. During this test all contacts shall remain in the de-energized position with zero coil voltage, and they shall remain in the energized position when the coil voltage Is reduced from the specified pickup voltage to one-half the specified voltage. A suitable indicating device shall be used to determine the ability of the relay contacts to remain in the proper position. Upon completion of this test the panel shall successfully complete the functional test, 4.6.10.

4.6.13 Vibration - The panel shall be subjected to the vibration test Procedure XII of MIL-E-5272 except that all tests shall be done at the room ambient temperature. During this test all contacts shall remain in the de-energlzed position with zero coil voltage, and they shall remain in the energized position when the coil voltage is reduced from the nominal voltage to the specified pickup voltage. A suitable indicating device shall be used to determine the ability of the relay contacts to remain In the proper position. Upon completion of this test the panel shall successfully complete the functional test, 4.6.10. Any structural, mechanical, or electrical failure shall be cause for rejection.

4.6.14 Shock - The panel shall be subjected to the shock test Procedure V of MIL-E-5272. During this test all contacts shall remain in the de-energized position with zero coil voltage, and all contacts shall remain In the energized position when the coil voltage is reduced from the nominal voltage to the specified pickup voltage. A suitable Indicating device shall be used to determine the ability of the relay contacts to remain in the proper position. Upon completion of this test the panel shall successfully complete the functional test, 4.6.10. Any structural, mechanical, or electrical failure shall be cause for rejection.

4.6.15 Salt Spray - The panel shall be subjected to salt spray test Procedure I of MIL-E-5272 for 50 hours. At the completion of the test the panel shall be cleaned by washing in cold running tap water and examined for deterioration. It shall then be dried for approximately 6 hours in a circulating air oven at approximately 57°C. Immediately after drying, it shall successfully complete the functional test, 4.6.10.

4.6.16 Humidity - The panel shall be subjected to the humidity test of Procedure I, MIL-E- 5272. At the completion of the test the panel shall successfully complete the functional test para. 4.6.10.

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4.6.17 Vibration Scan - The panel shall be subjected to the vibration test Method 204A, condition A of MIL-STD-202 except that only one cycle shall be performed in each of the three mutually perpendicular axes. The panel shall be subjected to one vibration scan in the de-energized condition and one scan in the energized condition. Coil voltage shall be the specified pick-up voltage. A suitable indicating device shall be used to determine the ability of the relay contacts to remain in the proper position. Contact chatter in excess of 10 microseconds shall constitute failure. Upon completion of this test the panel shall successfully complete the functional. test, 4.6.10. Any structural, mechanical, or electrical failure shall be cause for rejection.

4.6.18 Fungus - The panel shall be subjected to the fungus test procedure I of MIL-E-5272. Upon completion of this test the panel shall successfully complete the functional test, 4.6.10.

4.6.19 Flight Test - The panel shall be installed in an F4 aircraft having an electric power system conforming to MS90302 and shall accumulate 100 hours operating flight time without failure. This test will be performed by the government.

5. PREPARATION FOR DELIVERY

5.1 Preservation. Packaging and Packing- The panels shall be preserved and packaged in accordance with level A, B, or C of MIL-E-17555 as specified (6.2).

5.2 Marking and Labeling - Marking shall be as specified MIL-STD-129. Each container shall also be marked with the following information:

(a) Name of Unit
(b) Federal Stock Number
(c) Manufacturer's Part No.
(d) Manufacturer's Serial No.
(e) Military Part Number
(f) Contract or Order No.
(g) Date of Manufacture
(h) Quantity
(i) Name of Manufacturer
(j) Name of contractor (if different from Manuacturer)

6. NOTES

6.1 Intended Use - The panels covered by this specification are intended for use in aircraft as a means of remotely controlling the making and breaking of main electric power circuits.

6.2 Ordering Data - Procurement documents should specify the following:

(a) Specification Title and Number
(b) Military Part Number
(c) The packaging and packing level required (See 5.1)

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6.3 Qualification - Contract awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the Qualified Products List (QPL)。The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Government tested for qualification In order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for qualification of the panels manufactured in accordance with this specification is the Chief, Bureau of Naval Weapons, Departmnt of the Navy, Washington, D.C. 20360. Information pertaining to qualification testing and authorization to submit products for qualification may be obtained from the Commanding Officer, Naval Air Development Center, Johnsville, Warminster, Pennsylvania, 18974, Attention: AEED.

6.4 Definitions - The followlng terminology will be used in this specification:

6.4.1 Nominal Voltage - The standard designation which approximates the normal voltage of the system on which the panel is designed to operate (1.1).

6.4.2 System Voltage - The normal voltage of the electric system in which the panel will be required to operate (4.5.4).

6.4.3 Control Voltage - The voltage applied to the panel to energize the relay or contactor coils.

6.4.3.1 Maximum Control Voltage - The voltage which may be applied continuously (or under a specified operating cycle for Intermittent ratings) under the specified ambient conditions.

6.4.4 Pickup Voltage - The maximum voltage required to establish the energized function and above which the function remains established.

6.4.5 Drop-out Voltage - The maximum voltage at which the relay or contactor contacts will revert to the de-energized position.

6.4.6 Coil Current - The current which flows through the coil when maximum control voltage is applied with the contactor or relay stabilized at room temperature and the measurement taken within 10 seconds after application of coil voltage.

6.4.7 Operating Time - The elapsed time from the instant the coil is energized with nominal voltage to the moment all the normally closed contacts open or all the normally open contacts close, whichever is longer.

6.4.8 Release Time - The elapsed time from the moment the coil current starts to drop from its rated value to the moment all the normally open contacts open or the normally closed contacts close, whichever is longer.

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6.4.9 Contact Bounce - The time duration of uncontrolled making and breaking of contacts when contacts are moved to the closed position or when closed contacts are opened momentarily by dynamic forces.

6.4.10 Contact Disturbance - Any voltage that occurs across a pair of closed contacts caused by titration disturbance that is greater than 1 percent of the open circuit voltage.

6.4.11 Simultaneous Operation - Simultaneous operation time is defined as the period of time from the instant the first contact operates (either main or auxillary) until all contacts (either main or auxiliary) complete their operation including any contact bounce which may occur. This requirement applies for both pickup and dropout of the contactor or relay.

6.4.12 Qualification Samples - It is to be understood that upon receipt of the Letter of Authorization samples shall be furnished at no cost to the Government and that the manufacturer shall pay the transportation charges to and from the designated point where tests are to be made. In case of failure of the sample or samples submitted, consideration will be given to the request of the manucturer for additional tests only after it has been clearly shown that changes have been made in the product which the Qualifying Activity considers sufficient to warrant additional tests.

6.5 It is to be understood that panels supplied under contract shall be identical in every respect to the sample tested and found satisfactory, except for changes previously approved by the Government. Any unapproved changes from the qualification sample shall constitute cause for rejection.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication *or* otherwise as in any manner licensing the holder *or* any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-P-81305(WP) POWER DISTRIBUTION PANEL ALTERNATING CURRENT, F-4 AC SPEC FOR

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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